Top Pretrained Models for Transfer Learning

This document lists important pretrained models across different domains, their use cases, and where to access them.

Category	Model	Use Case	Where to Get
Computer Vision	ResNet	General Image classification	Hugging Face / PyTorch Hub
Computer Vision	VGG16 / VGG19	Simple deep CNN baseline	TensorFlow Hub
Computer Vision	EfficientNet	High performance & efficiency	TensorFlow Hub / PyTorch
Computer Vision	MobileNet	Lightweight (mobile apps)	TensorFlow Hub
Computer Vision	Vision Transformer (ViT)	Transformer-based vision tasks	Hugging Face
Object Detection	YOLO (v5/v8)	Real-time object detection	PyTorch / Ultralytics
Object Detection	Faster R-CNN	High accuracy detection	TensorFlow Hub
Object Detection	SSD (Single Shot Detector)	Fast object detection	TensorFlow Hub
NLP	BERT	Text classification, QA	Hugging Face
NLP	RoBERTa	Improved BERT	Hugging Face
NLP	GPT (GPT-2/3 style)	Text generation	Hugging Face
NLP	T5	Text-to-text tasks	Hugging Face
NLP	DistilBERT	Lightweight BERT	Hugging Face
Speech/Audio	Wav2Vec 2.0	Speech recognition	Hugging Face
Speech/Audio	DeepSpeech	Speech-to-text	GitHub
Generative Models	StyleGAN (StyleGAN)	Image generation	PapersWithCode
Generative Models	Stable Diffusion	Text to image generation	Hugging Face
Multimodal (Text+Image)	CLIP	Image-text matching	Hugging Face
Multimodal (Text+Image)	BLIP	Image captioning	Hugging Face
Tabular/General ML (Less DL)	XGBoost (pretrained possible)	Structured data tasks	Kaggle
Tabular/General ML (Less DL)	AutoML models	Automated ML solutions	Google AutoML